



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 6 Tissue Repair

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Learning Objectives

- ☐ Know what tissue repair is and how it occurs
- ☐ Know the types of tissue according to proliferative capacity .
- ☐ Know what healing by primary and secondary intention means.
- ☐ Know the factors that affect wound healing and the possible complications of wound healing.

Introduction-Tissue

Tissue Repair: Means the restoration of tissue architecture and function after an injury.

Repair

It occurs by two processes:

- 1.** Some tissues are able to replace the damaged cells and it return to a **normal state**; this process is called **regeneration**.
- 2.** If the injured tissues are unable to regenerate, or if the tissue severely damaged, **repair occurs by deposition of connective (fibrous) tissue**, a process termed **healing** that results in scar formation.

Mechanism of Repair

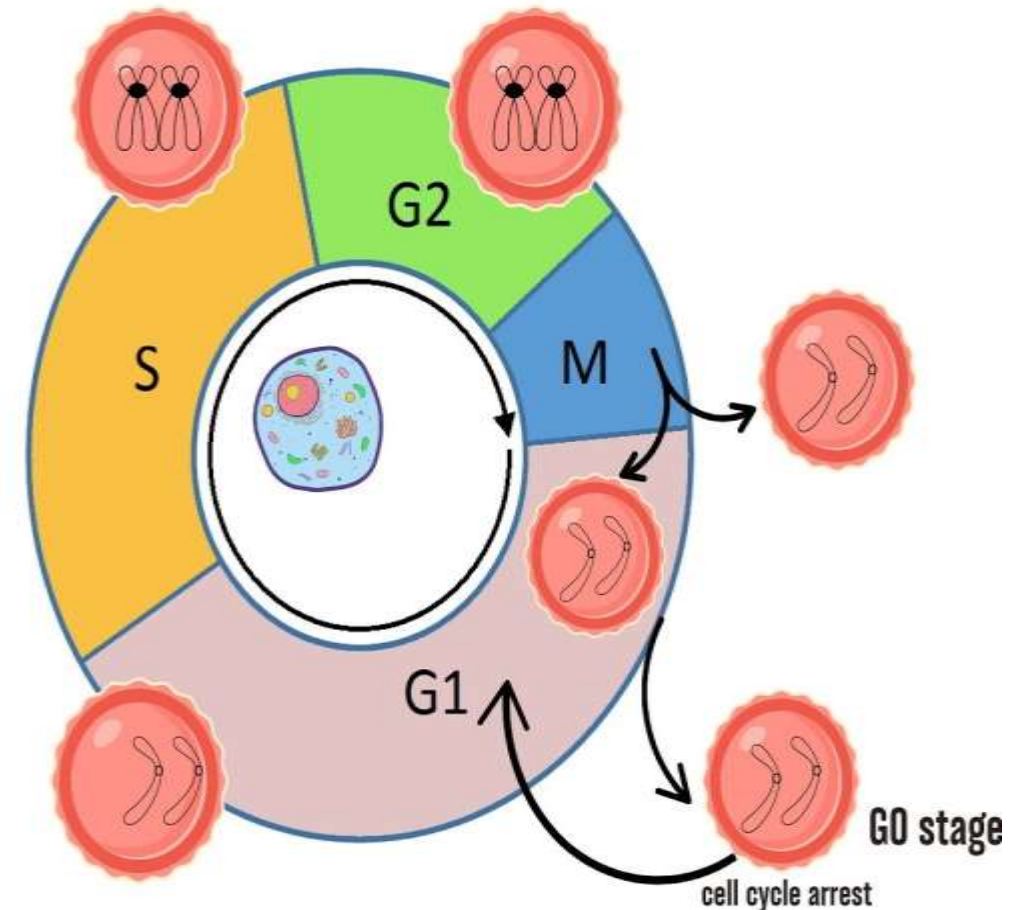
Repair involves the proliferation of various cells, The proliferation of these cell types is done by proteins that are collectively called **growth factors**.

Cell proliferation can be either:

- Physiologic (e.g. hormonal).
- Pathologic (e.g. injury, mechanical forces).

Non-dividing cells are either in cell cycle arrest in **G1** or they exit the cycle to enter a phase called **G0**

exposure to growth factors, needs to promote the G0/G1 transition and the entry of cells into the G1 phase of the cycle.



Groups of Tissue According to Proliferative Capacity

Group	Proliferative Status	Cell Cycle Position	Key Characteristics	Repair Mechanism	Examples
I. Labile Tissues	Continuously Dividing	Actively cycling, bypassing or having a very short G1 phase; never enter G0.	Cells are constantly lost and replaced.	Ready and Complete Regeneration	Hematopoietic cells, surface epithelia (skin, GI tract, oral cavity, urinary tract).
II. Stable Tissues	Resting	Normally reside in the G0 (resting) phase. Can move to G1 and re-enter the cycle upon stimulation	not dividing unless injured.	Regeneration (if little or mild), or Scar Formation (if severe).	Solid organs (liver, kidney, pancreas),
III. Permanent Tissues	Non-Proliferative	Terminally differentiated, irreversibly exited the cycle into a G0-like state	Cannot undergo mitosis in postnatal life.	Scar Formation (Irreversible loss of functional tissue).	Neurons, cardiac muscle cells, skeletal muscle.

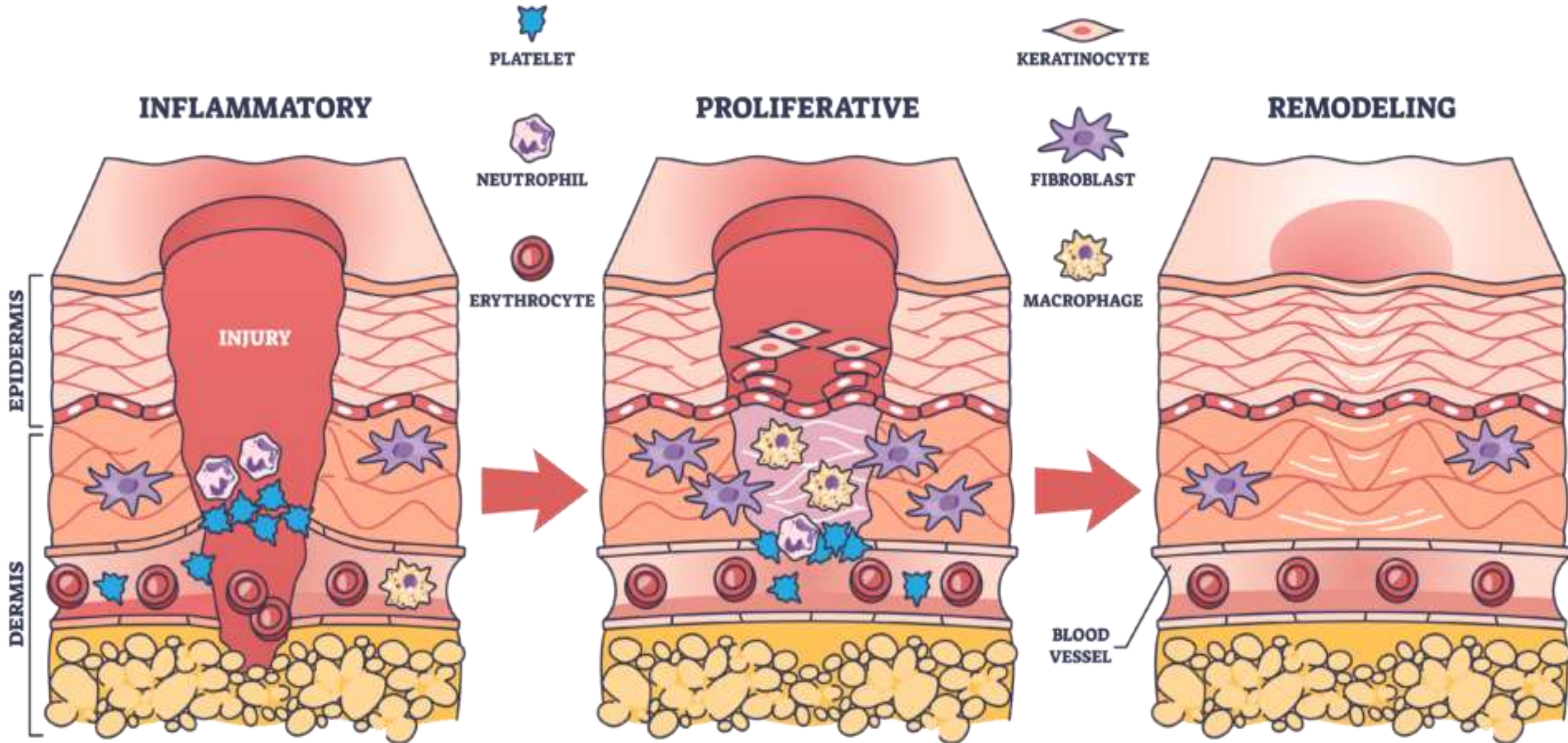
Wound Healing

Cutaneous wound healing has three main phases:

- (1)** inflammation,
- (2)** formation of granulation tissue,
- (3)** ECM deposition and remodeling.

Based on the **nature of the wound**, the healing of cutaneous wounds can occur by **first** or **second intention**

WOUND HEALING

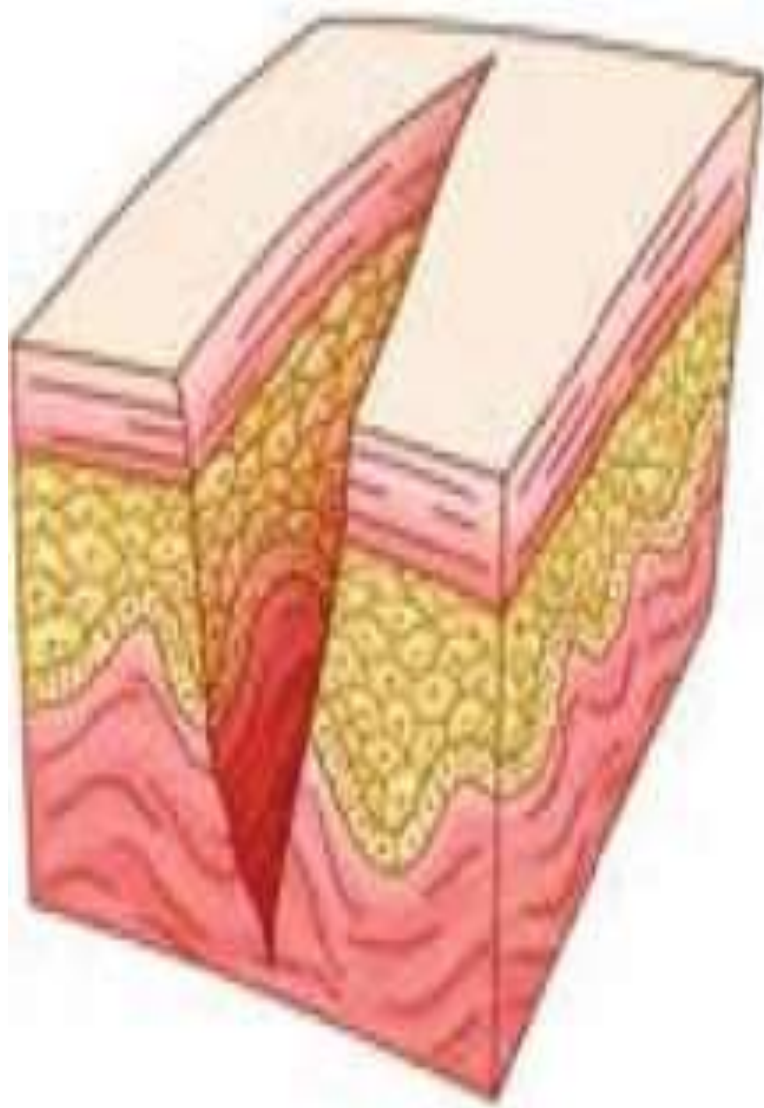


Healing by First Intention

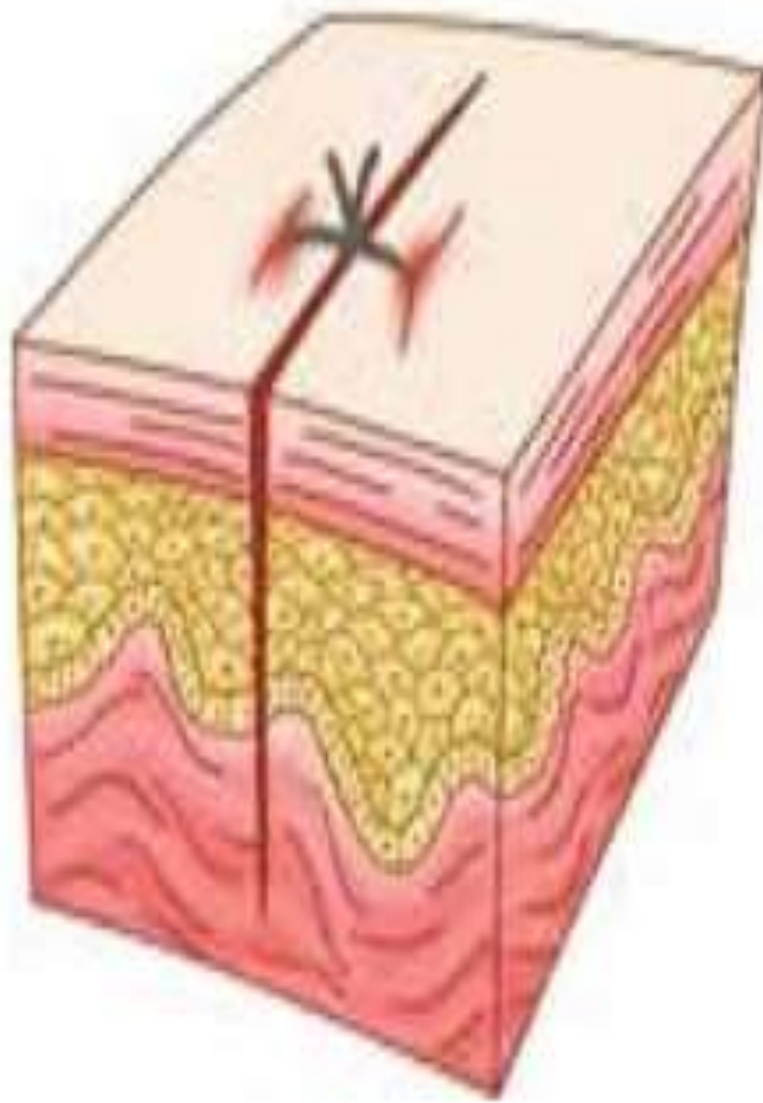
(Primary Union)

This is the ideal and simplest way a wound heals.

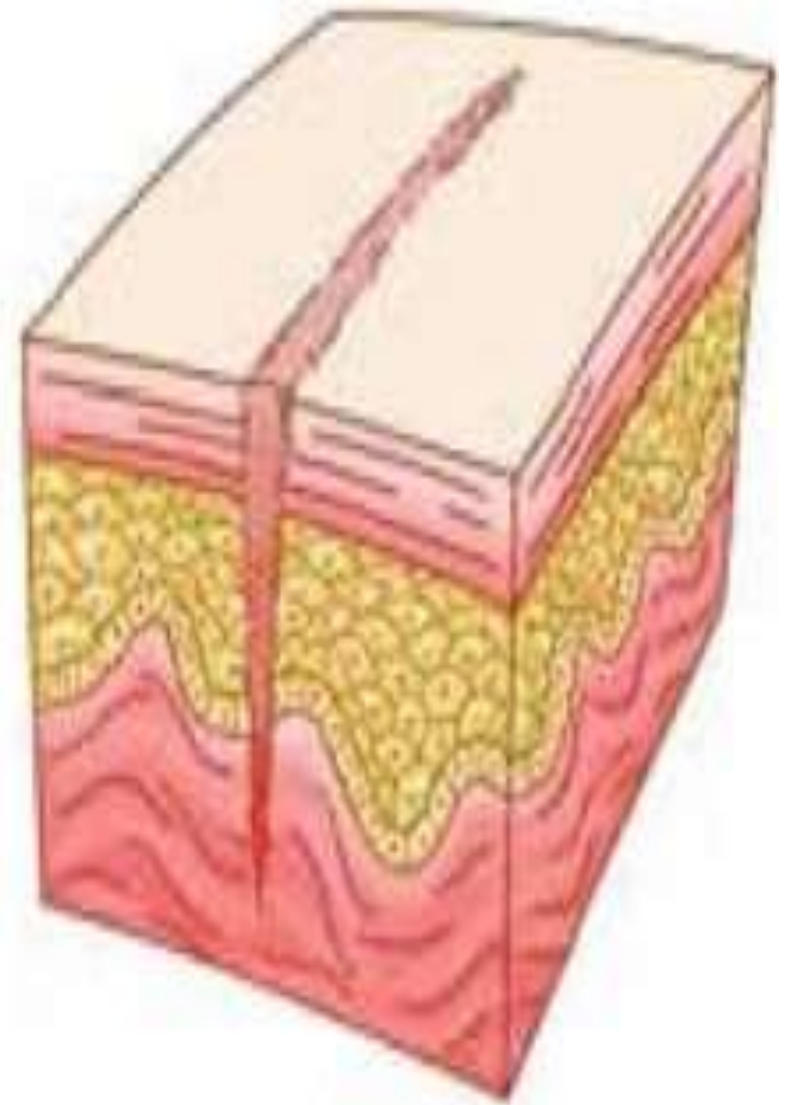
- ❑ **Wound Type:** A clean cut with minimal tissue damage (like a surgical incision or a sharp, small cut).
- ❑ **Wound Edges:** The edges are brought together and held in close contact (usually with sutures, staples, or glue).
- ❑ **Repair Process:** The body only needs to bridge a very narrow gap. Healing is dominated by epithelial regeneration (new skin growing across) with minimal connective tissue (scar) formation.
- ❑ **Outcome:** Fast healing and a fine, hairline scar



Cleaning the wound



Early suturing



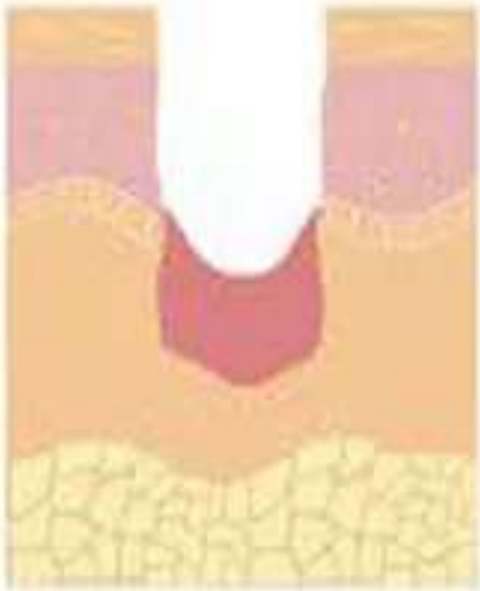
Linear scar

Healing by Second Intention (Secondary Union)

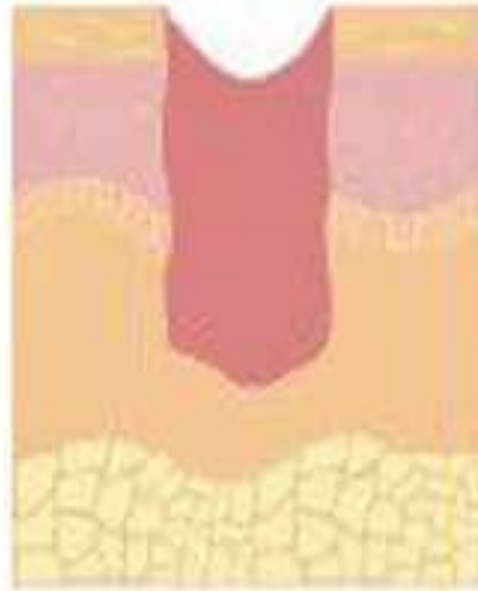
This occurs when there is a significant amount of missing tissue.

- ❑ **Wound Type:** A large, gaping wound, or one with extensive tissue loss, infection, or necrosis (like a deep burn, large ulcer, or abscess).
- ❑ **Wound Edges:** The edges cannot be pulled together; the wound must remain open.
- ❑ **Repair Process:** The entire wound cavity must first be filled from the bottom up with a mass of granulation tissue (new capillaries and collagen).
- ❑ **Outcome:** Slow healing and a large, visible scar (often irregular or indented) because of the sheer volume of new connective tissue laid down.

Secondary intention



Gaping irregular
wound



Granulation

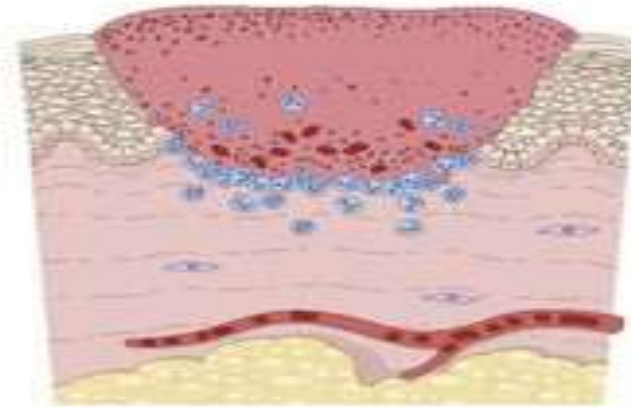
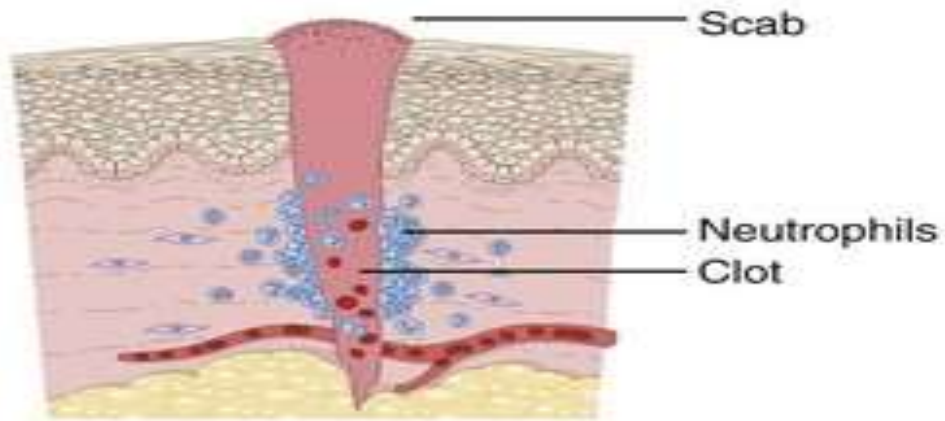


Epithelium grows
over scar

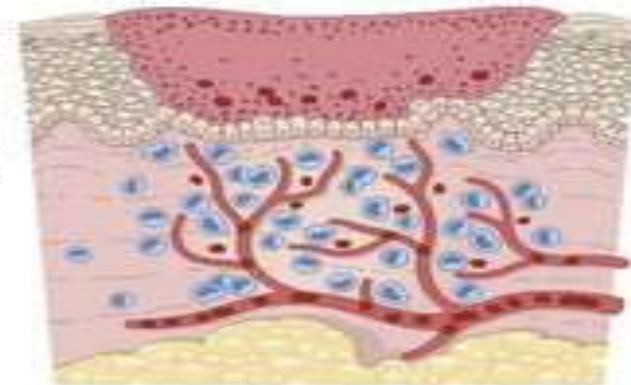
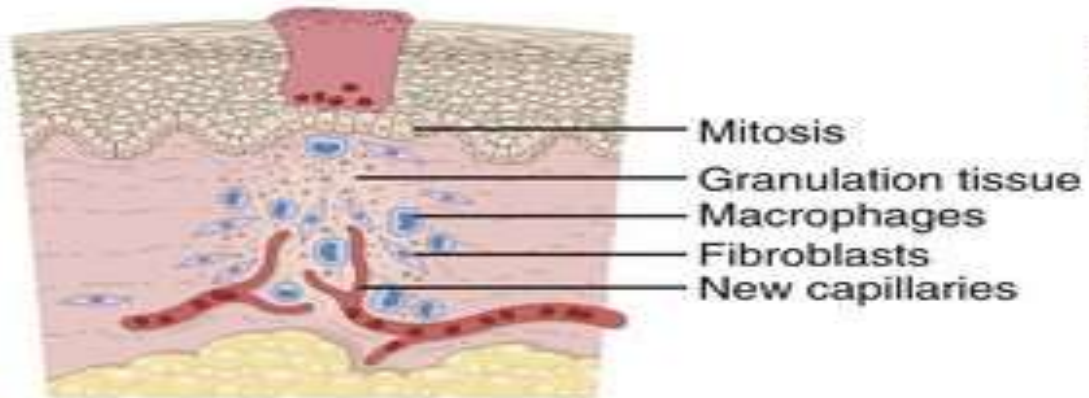
HEALING BY FIRST INTENTION

HEALING BY SECOND INTENTION

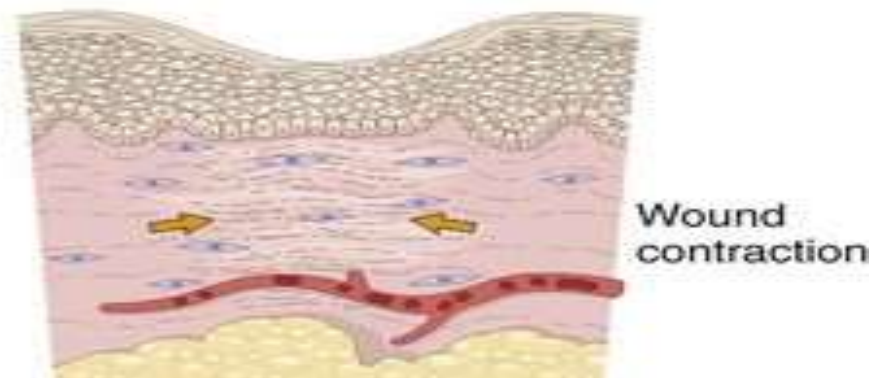
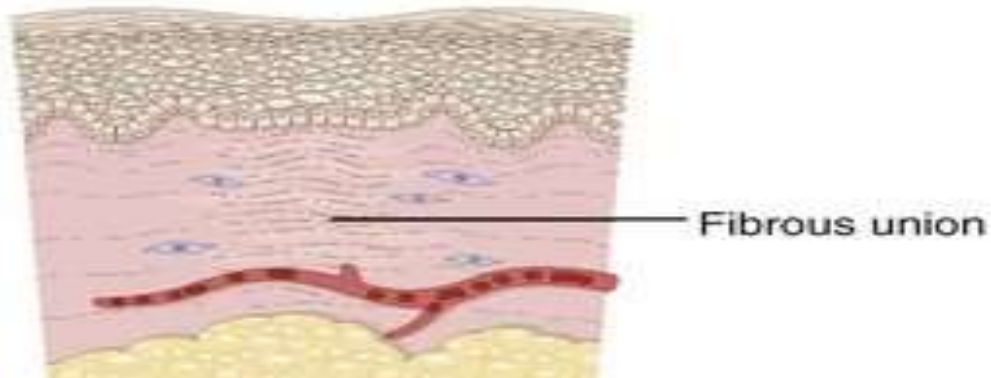
24 hours



3 to 7 days



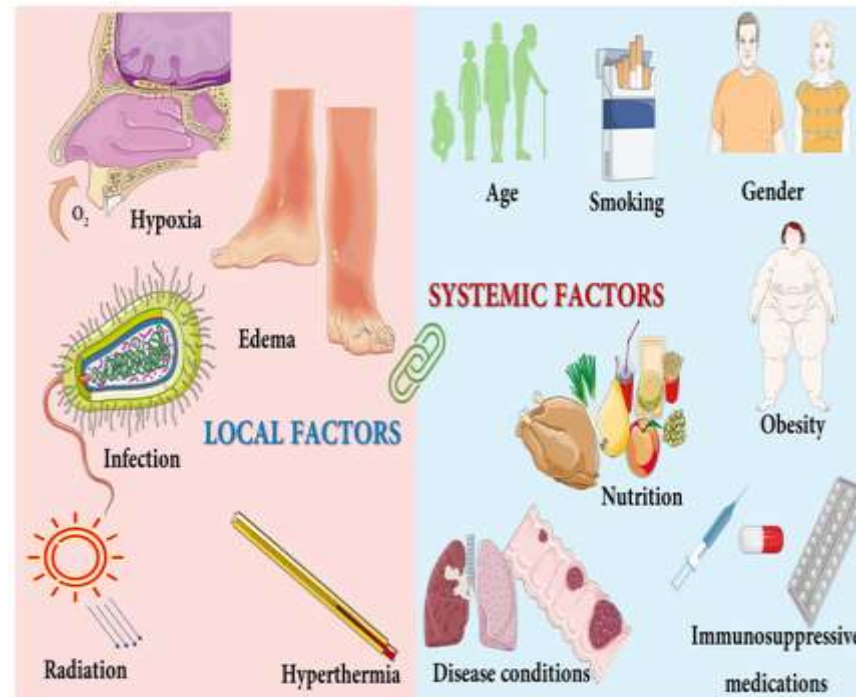
Weeks



Factors that influence wound healing

1-Local Factors:

- Infection (single most important reason for delayed wound healing)
- Foreign bodies
suture material, bone
- Mechanical factors
Early movement



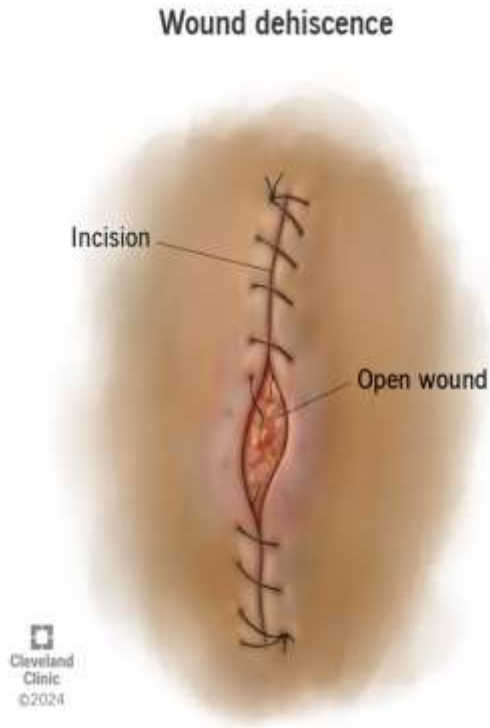
2-Systemic factors:

- Malnutrition
- Metabolic status
Diabetes mellitus
- Circulatory status
- Inadequate blood supply

Complications of wound healing

1. Deficient scar formation

- Wound dehiscence



2. Excessive formation of scar tissue

- Keloid (excessive collagen deposition)



3. contractures

